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Complete List of Authors:	Tang, Jie; Xihua University, School of Food and Biotechnology Hu, Qiong; Xihua University Liu, Bo; Xihua University Lei, Dan; Xihua University Chen, Ting; Xihua University Sun, Qing; Xihua University Zeng, Chao; Xihua University Zhang, Qing; Xihua University
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**Efficient biodegradation of 3-phenoxybenzoic acid and pyrethroid
pesticides by a novel strain *Klebsiella pneumoniae* BPBA052**

Jie Tang*, Qiong Hu, Bo Liu, Dan Lei, Tingting Chen, Qing Sun, Chaoyi Zeng, Qing Zhang

Key Laboratory of Food Biotechnology, School of Food and Biotechnology, Xihua University, 610039
Chengdu, Sichuan, PR China

*Corresponding author:

Prof. Jie Tang

Tel./Fax: +86 28 87720552

Email: wendyjiejie@tom.com

Draft

Abstract

A novel 3-phenoxybenzoic acid (3-PBA)-degrading strain (BPBA052) of *Klebsiella pneumoniae* was isolated from soybean rhizosphere soil. The strain was screened by enrichment domestication, isolation, and purification, using 3-PBA as the sole carbon and energy source. It could degrade 96.37% of 3-PBA (100 mg/L) within 72 h, and its growth and 3-PBA degradation followed logistic growth ($X_{BPBA052} = 0.0883e^{0.0947t}/[1-0.0792 \times \{1-0.0883e^{0.0947t}\}]$; $\mu_m = 0.0947 \text{ h}^{-1}$, $X_0 = 0.0883$, and $X_m = 1.1145$) and first-order degradation ($C_{BPBA052} = 101.8194 \times e^{-0.0403t}$, $k = 0.0403$, $t_{1/2} = 17.22 \text{ h}$) kinetic models, respectively. Based on Box–Behnken-response surface analysis, the optimal temperature, pH and 3-PBA concentration for *K. pneumoniae* BPBA052 were 35.01 °C, 7.77, and 150 mg/L. Moreover, PPs (such as beta-cypermethrin, permethrin, bifenthrin, deltamethrin, and fenvalerate) and 3-PBA metabolites (including phenol, catechol, and protocatechuate) were efficiently utilized by BPBA052. We propose a novel microbial metabolic pathway for 3-PBA, based on metabolite identification; enzyme-degrading activity; and cloning of the phenol hydroxylase, catechol 1,2-dioxygenase, and protocatechuate 3,4-dioxygenase genes. This study provides a fundamental platform for further studies to reveal the 3-PBA-biodegradation mechanism and show *K. pneumoniae* BPBA052 as a potential microbial resource for bioremediation of environments polluted with 3-PBA or PPs.

Keywords: 3-phenoxybenzoic acid, pyrethroid pesticides, *Klebsiella pneumoniae*, biodegradation, metabolic pathway

1. Introduction

Pyrethroid pesticides (PPs), a class of synthetic bionic insecticides, are widely used in protecting fruits, vegetables, tea, and cereals from pests, as well as mosquitoes, cockroaches, and livestock parasites, owing to their high efficiency and low toxicity (Cycoń and Piotrowska-Seget 2016; Watkins et al. 2016; Ye et al. 2016). However, most PPs residues are easily enriched in environment and agricultural products, and finally accumulate in animals and humans through the food chain (Watkins et al. 2016; Ye et al. 2016). Data from many studies have shown that PPs lead to highly acute toxicity in aquatic organisms and invertebrates, and their neurotoxicity, immunotoxicity, and reproductive toxicity affect humans and animals (Zhang et al. 2010). Moreover, 3-phenoxybenzoic acid (3-PBA), the main metabolite of PPs, causes significant levels of secondary pollution, as compared to the levels of direct pollution and toxic effects caused by PPs in the environment and human body (McCoy et al. 2012). Recent findings have shown that 3-PBA is an endocrine disruptor with immunotoxic effects (Wang et al. 2017), exhibits estrogenic and anti-androgenic activity in mammals (Jin et al. 2010; Sun et al. 2014). In addition, 3-PBA was exceedingly recalcitrant to microbial degradation having a half-life of up to 180 d (Martínez et al. 2009), and exerted a toxic inhibitory effect that prevented further PPs degradation (Halden et al. 1999). Therefore, the accumulation and long-term use of 3-PBA has caused great concern regarding environmental safety and public health. The elimination of 3-PBA residues is crucial.

At present, the remediation of environmental pollution mainly relies on physical catalysis, chemical oxidation, and bioremediation. Among these strategies, bioremediation is based on the catabolic activity of living microorganisms for pollutant elimination and has emerged as the most promising, eco-friendly, efficient, and cost-effective approach (Cycoń and Piotrowska-Seget 2016). Therefore, screening for efficient 3-PBA-degrading strains is a primary task for eliminating 3-PBA and PPs residues from the environment. Some bacteria such as *Sphingomonas* sp. (Tang et al. 2013), *Pseudomonas* sp. (Halden et al. 1999), *Ochrobactrum*

lupine (Chen et al. 2011a), *Sphingobium jiangsuense* (Zhang et al. 2012), and *Stenotrophomonas* sp (Chen et al. 2011b) have been isolated from active sludge. In addition, *Bacillus* sp (Chen et al. 2012a). and *Achromobacter* sp. P-01 (Chen et al. 2011c) were isolated from pesticide-contaminated soil, and *Aspergillus oryzae* (Zhu et al. 2016) was isolated from soy sauce. However, strains that efficiently degrade 3-PBA under high concentration stress have rarely been reported. Furthermore, studies on microorganisms capable of degrading not only 3-PBA but also its parent PPs are still rare.

In this study, we isolated and evaluated a novel strain of *K. pneumoniae* (BPBA052) from soybean rhizosphere soil, which exhibited efficient 3-PBA- and PPs-degradation capacities. Optimization of 3-PBA biodegradation and the degradation-kinetic models of *K. pneumoniae* BPBA052 were also investigated and established. In addition, this report contains the first proposal of a microbial degradation pathway for 3-PBA, based on metabolite identification, enzyme-degrading assays, and cloning of enzyme-degrading genes. Collectively, our findings provide a theoretical basis for the further bioremediation of 3-PBA or PPs that polluted the environment.

2. Materials and methods

2.1. Chemicals and media

The 3-PBA standard (98.0%) and chromatographic-grade acetonitrile were purchased from TCI (Shanghai, China) and Adamas-Beta (Shanghai, China), respectively. Beta-cypermethrin (96.8%), permethrin (95.2%), bifenthrin (97.1%), fenvalerate (96.0%), and deltamethrin (98.5%) were obtained from Ronch Chemicals (Nanjing, China). Plasmid pGEM®-T and *E. coli* DH5α (Promega Biotechnology Co. Ltd., China) were used for cloning purposes. A Gel Extraction Kit and a Plasmid Mini Kit were purchased from Omega Bio-tek (Norcross, GA), and T4 DNA ligase and restriction endonucleases were obtained from TaKaRa (Otsu, Japan). All other

chemical reagents were commercially available.

Liquid and solid Luria–Bertani (LB) medium and mineral salt medium (MSM) were prepared as described previously (Tang et al. 2018).

2.2. Enrichment, isolation, and screening of bacterial strains

An enrichment-culture technique was used to isolate degrading strains. Soil samples were collected from soybean rhizosphere contaminated with PPs in Northern Sichuan, China. Initial culture enrichment was performed in a 250-mL Erlenmeyer flask containing 50 mL sterilized enrichment medium with 1 g of soil sample and 25 mg/L of 3-PBA, which was incubated at 30 °C and 180 rpm for 5 days. Subsequently, 5 mL medium was transferred to 50 mL of fresh enrichment medium, six additional successive transfers were performed with 3-PBA concentrations of 50, 100, 200, 400, 800, and 1600 mg/L; the final culture was serially diluted, plated, and incubated on MSM plates containing 100 mg/L 3-PBA (Tang et al. 2018). The ability of the purified strains to degrade 3-PBA was measured by high-performance liquid chromatography (HPLC), and one efficient strain (BPBA052) was selected for further studies. The 3-PBA-degradation rate was calculated according to Eq. (1), where C and C_0 represented the 3-PBA content in the inoculated medium and control medium, respectively.

$$\text{Degradation rate (\%)} = (1 - C/C_0) \times 100\% \quad (1)$$

2.3. Taxonomic identification of the BPBA052 strain

To characterize strain BPBA052, we analyzed its morphology, physio-biochemical characteristics, and 16S rRNA gene sequence. Cell morphology was observed using a scanning electron microscope (Japan Electronics Co., Ltd., Japan). Physio-biochemical tests were performed as described in Bergey's Manual of Determinative

Bacteriology (Brenner and Krieg 2005). Genomic DNA extracted from a bacterial suspension that was cultured in LB medium at 30 °C for 24 h and was used as the PCR template. Genomic DNA was extracted and the 16S rRNA gene was amplified, as described previously (Tang et al. 2018). The Gel Extraction Kit was used to recover the amplified 16S rRNA. Transformants were selected by blue-white colony screening. Plasmid DNA was extracted using the Plasmid Mini Kit and sent to Tsingke Biological Technology Co. (Chengdu, China) for sequencing. The sequence was compared with sequences available in GenBank using the NCBI BLAST tool (<http://www.ncbi.nlm.nih.gov/blast/>) for identification. MEGA 7.0 software was used to construct a phylogenetic tree.

2.4. Inoculum preparation

BPBA052 was thawed and inoculated on LB plates at 30 °C for 24 h. The isolate was transferred to LB medium, harvested by centrifugation at 6300 g force for 2 min, washed twice with 0.9% sterile N-saline, and adjusted to an optical density of approximately 1.0 at 600 nm to prepare the inocula. One percent of this suspension (approximately 1.0×10^8 cells/mL) was used as the inoculum for 3-PBA- and PPs-biodegradation studies.

2.5. Optimal conditions for 3-PBA degradation with strain BPBA052

The single-factor test was designed to optimize the conditions for 3-PBA degradation by BPBA052 under the following parameters: initial pH (5.5–9.5), temperature (25–40 °C), and 3-PBA concentration (25–500 mg/L). Subsequently based on single-factor tests, a Box–Behnken design consisting of 17 experiments (Table 1) was used to build quadratic models, which were generated with Design-Expert software (version 10.0 Stat-Ease Inc., Minneapolis, USA). Each treatment was performed in triplicate to estimate the experimental error, and

non-inoculated cultures served as controls. Data were processed using the response surface-regression procedure to fit the quadratic polynomial equation (Eq. 2), where X_i and X_j are independent variables (X_1 temperature, X_2 3-PBA concentration, X_3 initial pH), Y_i represents the predicted response, β_0 represents a constant, β_i represents the linear coefficient, β_{ij} represents the interaction coefficient, and β_{ii} represents the quadratic coefficient.

$$Y_i = \beta_0 + \sum_{i=1}^k \beta_i X_i + \sum_{i=1}^{j-1} \sum_{j=1}^n \beta_{ij} X_i X_j + \sum_{i=1}^k \beta_{ii} X_i^2 \quad (2)$$

2.6. Degradation kinetics of various PPs, determined using the BPBA052 strain

To further study its ability to degrade various PPs, the BPBA052 strain was inoculated in liquid medium supplemented with beta-cypermethrin, permethrin, bifenthrin, deltamethrin, and fenvalerate (100 mg/L each), and incubated under optimal degradation conditions for 48 h. All experiments were performed in triplicate to estimate the experimental error, and non-inoculated cultures served as controls. The samples were collected and analyzed at an interval of 8 h, and the pesticide residues were measured by HPLC.

2.7. HPLC conditions and analysis

Analysis of sample residues in filtrates was performed using a ZORBAX eclipse plus C18 column (4.6 mm $\times$ 150 mm, 5 μ m) with an Agilent 1100 HPLC system. PPs were analyzed using a mobile phase consisting of ultrapure water and acetonitrile in a 10:90 ratio (v/v), at a flow rate of 1 mL/min. In addition, 3-PBA was analyzed by HPLC using a mobile phase consisting of water and acetonitrile in a 30:70 ratio (v/v), a flow rate of 1 mL/min, and a water pH adjusted to 2.0 using phosphoric acid. PPs and 3-PBA were detected at 210 nm.

2.8. 3-PBA metabolites and the enzyme assay

The inoculum (5.0% v/v) was added to 20 mL LB medium containing 3-PBA (150 mg/L) in a 100 mL

Erlenmeyer flask and incubated under optimal degradation conditions for 8, 16, 24, 32, 40, 48 h. Metabolite detection was based on a previous GC-MS study by Tang et al. (2013). The ability of BPBA052 to degrade potential intermediate metabolites (catechol, phenol, benzoic acid, phthalate, procatechuate) were measured by HPLC under optimal degradation conditions (Chen et al. 2012a; Tang et al. 2013). Preparation of the enzyme solution and determining the subcellular location of the enzyme were performed as previously described (Moreno et al. 2011; Dalal et al. 2012). Determination of the enzymatic activity of phenol hydroxylase (LPH) and protocatechuate 3,4-dioxygenase (PRO) (Maloney et al. 1988; Ngai et al. 1990; Bradford MM 1976), and the assays for catechol 1, 2-dioxygenase (CatA) and catechol 2, 3-dioxygenase (CatB) (Santos et al. 2009; Milo et al. 1999) were performed as described in previous studies. The activity unit (U) of a substrate-degrading enzyme was defined as the amount of enzyme required to reduce the amount of substrate by 1.0 nmol within 1.0 min.

2.9. Degrading-enzyme gene cloning and analysis

The core regions of enzyme-degrading genes, including *LPH*, *CatA*, *CatB*, *PRO* subunit alpha (*PRO α*), and subunit beta (*PRO β*), were amplified using degenerate primers (Supplementary Table 1), which were designed with DNAMAN software version 8.0. Gene sequencing was performed as described in the above procedure (taxonomic identification of the BPBA052 strain). The BLASTx and BLASTp programs (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>) were used to analyze the nucleotide and amino acid sequences, respectively. Softberry software (<http://linux1.softberry.com>) and SignalP 4.0 (<http://www.cbs.dtu.dk/services/SignalP/>) were used to predict the introns and signal peptide, respectively. Vector NTI Advance 10.0 software was used to predict the molecular weight and isoelectric value.

2.10. Statistical analysis

Statistical analysis was performed using Origin software (version 8.5). Each experiment was performed in triplicate to estimate the experimental error, with no inoculation used as a control, and the values are presented as the mean and standard deviation.

3. Results

3.1. Isolation and characterization of the BPBA052 strain

Owing to enrichment and isolation, 7 morphologically different pure isolates (grown in the presence of 3-PBA at concentrations of up to 1600 mg/L) were obtained from the enrichment culture medium. Our results showed that strain BPBA052 exhibited a high 3-PBA-degradation rate and degraded 96.37% of 3-PBA (100 mg/L) within 72 h. hence, it was selected for further studies. BPBA052 is a gram-negative, facultative anaerobic, short, and rod-shaped bacterium that is 0.6 to 1.1 μm in length and 0.3 to 0.5 μm in width (Fig. 1a). Colonies grown on violet red bile agar plates are magenta colored, opaque, round, viscous, and central-convex shaped, with a white precipitate at the margin. BPBA052 tested positive in terms of nitrate reduction; the Voges–Proskauer test; hydrolyzed urea and lysine production; and inositol, glucose, lactose, and citrate utilization. It tested negative for methyl red, indole reaction, catalase, hydrogen sulfide production, hydrolyzed arginine, and ornithine. Phylogenetic analysis of the 16S rRNA gene sequence, which was conducted using BLAST analysis (Fig. 1b), revealed that the degree of homology of strain BPBA052 with *K. pneumoniae* CICC10281 (KF359886) and *K. pneumoniae* JCM1662 (NR113240) was 99%. Thus, BPBA052 was identified as a *K. pneumoniae* strain (KY425746, Collection number: CGMCC 7.307).

3.2. Utilization of 3-PBA for growth by *K. pneumoniae* BPBA052

Fig. 2 shows *K. pneumoniae* BPBA052 growth and 3-PBA degradation in MSM containing 100 mg/L 3-PBA as the sole carbon and energy source. The results showed that 3-PBA degradation correlated with the growth of *K. pneumoniae* BPBA052 and that >96% of the 3-PBA was degraded within 72 h. Models reflected the growth regularity and 3-PBA-degradation process of *K. pneumoniae* BPBA052. A cell growth-kinetics model (Eq. 3) and a first-order degradation-kinetic model (Eq. 4) were fit to the experimental values, resulting in the following equations: $X_{BPBA052} = 0.0883e^{0.0947t}/(1 - 0.0792 \times [1 - 0.0883e^{0.0947t}])$, where $\mu_m = 0.0947 \text{ h}^{-1}$, $X_0 = 0.0883$, and $X_m = 1.1145$; $C_{BPBA052} = 101.8194 \times e^{-0.0403t}$, where $k = 0.0403$ and $t_{1/2} = 17.22 \text{ h}$. Here, X represents the predicted 3-PBA degradation (%) caused by *K. pneumoniae* BPBA052, X_0 is the initial cell concentration (OD₆₀₀), X_m is the maximum cell concentration (OD₆₀₀), t is the culture time (h), and μ_m is the maximum specific growth rate (h⁻¹). C_0 represents the initial 3-PBA concentration (mg/L), C_t represents the 3-PBA concentration at time t (mg/L), t represents the degradation time (h), k represents the degradation rate constant (h⁻¹), and $t_{1/2}$ represents the half-life of 3-PBA.

$$X = X_0 e^{\mu_m t} / (1 - (X_0 / X_m)(1 - e^{\mu_m t})) \quad (3)$$

$$C_t = C_0 \times e^{-kt}, \quad t_{1/2} = \ln 2 / k \quad (4)$$

3.3. Optimization of 3-PBA degradation by strain *K. pneumoniae* BPBA052

The kinetics of 3-PBA degradation by *K. pneumoniae* BPBA052 at different culture temperatures, initial 3-PBA concentrations, and initial pH values are shown in Fig. 3. Different 3-PBA-degradation rates were observed when *K. pneumoniae* BPBA052 was cultured at five temperatures (25–40 °C; Fig. 3a). In cultures incubated at 34 °C and 37 °C, 3-PBA degradation started rapidly within 8 h with almost no lag phase, and >80% of the 3-PBA was degraded within 48 h. However, the degradation rates of 3-PBA were 53.6%, 75.8%, and 67.6% within 48 h at 25, 30, and 40 °C, respectively. At low 3-PBA concentrations (25–100 mg/L), the

degradation rate reached 80% within 48 h (Fig. 3b). Although at higher concentrations (200–500 mg/L), only 70% degradation was achieved within 48 h, the toxic inhibitory effect of 3-PBA on *K. pneumoniae* BPBA052 was not obvious. The pH is another important factor affecting 3-PBA degradation. Thus, we determined the effect of the pH on the ability of *K. pneumoniae* BPBA052 to degrade 3-PBA at different pH values of 5.5, 6.5, 7.5, 8.5, and 9.5 (Fig. 3c). The optimal pH value was found to be 7.5, where the degradation rate reached 83.8% within 48 h.

Based on the Box–Behnken design, three significant variables, including the culture temperature (X_1), 3-PBA concentration (X_2), and initial pH (X_3) were applied to optimize the biodegradation conditions using response surface methodology (Liu et al. 2014). Subsequently, the data from Table 1 were analyzed using the response surface-regression procedure, and the analysis of variance (ANOVA) results for the fitted quadratic polynomial model are presented in Table 2. A quadratic polynomial function was fit to the experimental values to obtain equation Eq. 5, where Y represents the predicted 3-PBA degradation (%) by *K. pneumoniae* BPBA052, and X_1 , X_2 , and X_3 are the coded values for the culture temperature, initial concentration, and initial pH, respectively.

$$Y = 84.73600 - 4.65250X_1 + 1.34875X_2 + 3.51375X_3 - 0.21250X_1X_2 - 2.29750X_1X_3 - 0.24500X_2X_3 - 4.10550X_1^2 - 0.013000X_2^2 - 8.97800X_3^2 \quad (5)$$

ANOVA and F testing were essential to evaluate the polynomial model and the relationship between the responses (the 3-PBA-degradation rate) and the variables. The R^2 value of 0.9870 indicated the adequacy of the model, which explained 98.70% of the response variability, implying that the predicted values of the model had a high degree of correlation with the experimental values. The accuracy of the model was also supported by the high value of the adjusted R^2 (0.9704). The lack-of-fit term was non-significant ($P > 0.05$) and showed that the quadratic model was sufficient for predicting 3-PBA degradation. Moreover, a low coefficient of variation value

(1.5%) also indicated a high degree of precision and reliability of the model. Therefore, the established accurate model was highly efficient for predicting and optimizing the 3-PBA-biodegradation conditions within the chosen parameters (culture temperature, 3-PBA concentration, and initial pH). Regression analysis revealed that the linear and square terms of culture temperature (X_1) and initial pH (X_3) were important factors ($P < 0.05$) in 3-PBA degradation by *K. pneumoniae* BPBA052. The linear term of the 3-PBA concentration (X_2) and the interaction terms played unimportant roles ($P > 0.05$) in 3-PBA degradation (Fig. 4).

As shown in Fig. 4, the theoretical maximum value of 3-PBA biodegradation at the stationary point was 88.13% within 48 h. Based on the established model, the optimal parameters for 3-PBA degradation by *K. pneumoniae* BPBA052 were determined, including a temperature, initial pH, and 3-PBA concentration of 35.01 °C, 7.77, and 150 mg/L, respectively. Results of the validation experiments showed that the average degradation rate under optimized conditions was 87.75% within 48 h, which was consistent with the predicted value. The results indicated that the model could accurately predict the 3-PBA-degradation rate by *K. pneumoniae* BPBA052 under different culture conditions.

3.4. Degradation kinetics of various PPs by the *K. pneumoniae* BPBA052 strain

To further quantify the degradation ability of *K. pneumoniae* BPBA052 against different PPs, a first-order model was used to express the kinetics (Chen et al. 2011a). *K. pneumoniae* BPBA052 was capable of degrading PPs, and degradation rates of up to 45.42%, 30.86%, 21.74%, 40.57%, and 28.41% were observed for different PPs such as beta-cypermethrin, permethrin, bifenthrin, deltamethrin, and fenvalerate, respectively, within 48 h. As shown in Table 3, the degradation process was characterized by a constant (k) ranging from 0.0067 to 0.0132 h⁻¹ with a degradation half-life ($t_{1/2}$) of 52.51 to 103.45 h. The R^2 values were calculated to be 0.9792–0.9965, indicating that the degradation rate fit well with the mathematical model.

3.5. Mechanism of 3-PBA biodegradation by the *K. pneumoniae* BPBA052 strain

The 3-PBA-degradation products were identified by gas chromatography-mass spectrometry (GC-MS). The results of GC-MS analysis showed that with the decrease of 3-PBA no other degradation products were generated after inoculation for 8, 16, 24, 32, 40, 48 h. Degradation products could have been rapidly mineralized by the enzyme-degradation system of *K. pneumoniae* BPBA052. Thus, we further evaluated the sustainable degradation capability of potential products and found that the degradation rates of speculative metabolites of 3-PBA by *K. pneumoniae* BPBA052 (including catechol, phenol, benzoic acid, phthalate, and protocatechuate) were 71.81, 96.32, 13.55, 5.55, and 57.61%, respectively, within 24 h (Fig. 5a). It indicated that it is difficult for *K. pneumoniae* BPBA052 to degrade benzoic acid and phthalate. So we proposed that phenol and protocatechuate were first formed by the oxidization and cleavage of the diaryl ether of 3-PBA (Tallur et al. 2008). Then, catechol was generated by the oxygenation of phenol and decarboxylation of protocatechuate. Catechol and protocatechuate rings were rapidly cleaved by the catalyzation of CatA and PRO, resulting in the formation of cis,cis-muconate and carboxy-cis,cis-muconate, which were metabolized by pyruvate metabolism to produce acetyl coA. Eventually, acetyl coA was converted into CO₂ and H₂O in the Krebs cycle, as shown in Fig. 5b (Chen et al. 2012b). Subsequently, the key enzymes LPH, Cat, and PRO, which have been associated with the production of metabolites, were also analyzed. The enzyme activities of LPH, CatA, CatB, and PRO were determined to be 23.45, 15.25, 3.42, and 19.11 U/mL, respectively. Thus, we studied the genes encoding the enzymes responsible for this degradation (which are involved in the downstream processing of 3-PBA biodegradation) and successfully cloned the *LPH*, *CatA*, and *PROβ* genes from the genome of *K. pneumoniae* BPBA052. Sequencing the inserts of plasmids containing the *LPH*, *CatA*, and *PROβ* genes (GenBank accession numbers MF987472–MF987474) revealed that the open reading frames of the *LPH*, *CatA*, and *PROβ* genes

encode proteins with 210, 305, and 245 amino acid residues and calculated molecular weights of 24.44, 33.25, and 28.81 kDa, respectively. From a molecular biological perspective, the cloning of *LPH*, *CatA*, and *PROβ* further demonstrated that *K. pneumoniae* BPBA052 can degrade metabolites, and further verified the pathway for 3-PBA.

4. Discussion

3-PBA not only cause serious environmental contamination, but also raised public concern about human health around the world (Chen et al. 2012a; Zhao et al. 2016). Therefore, an eco-friendly and efficient treatment for the threats of 3-PBA pollution is extremely urgent, and bioremediation is considered the most promising approach. Regrettably, attempts to isolate bacteria that are resistant to high concentration of 3-PBA are usually unsuccessful, owing to the antimicrobial property of 3-PBA (Chen et al. 2011a; Chen et al. 2011b); and most microbes can only degrade 3-PBA but not to degrade PPs. With this aim, we successfully screened for a novel strain. By determining its morphological and physio-biochemical characteristics, and performing 16S rRNA gene sequencing analysis, we identified a desirable strain of *K. pneumoniae* designated here as BPBA052. Although studies have shown that *K. pneumoniae* mostly has a biological nitrogen-fixation function (Kuan et al. 2016), Abraham and Silambarasan reported that *Klebsiella* bacteria have notably broad catabolic capabilities for mineralizing endosulfan and its metabolite (Abraham and Silambarasan et al. 2014). Our results showed that *K. pneumoniae* BPBA052 can be applied for bioremediating 3-PBA or PPs, and it possessed an efficient 3-PBA-degradation when the concentration was up to 1600 mg/L. Using 100 mg/L 3-PBA as the sole carbon and energy source, >96% of the 3-PBA was degraded by *K. pneumoniae* BPBA052 within 72 h. Zhu et al. (2016) reported that *Aspergillus oryzae* M-4 could degrade 80.62% of 3-PBA (100 mg/L) within 5 d; and Chen et al. (2012a) reported that *Bacillus* sp. strain DG-02 could degrade 95.6% of 3-PBA (50 mg/L) within 72 h; While *K.*

pneumoniae BPBA052 exhibited a higher degradation rate than the *Aspergillus oryzae* M-4 and *Bacillus* sp. DG-02 strains. This study is the first to provide evidence that *Klebsiella* bacteria can participate in 3-PBA biodegradation and expands the diversity of 3-PBA-degrading strains.

It is noteworthy that *K. pneumoniae* BPBA052 could not only degrade and utilize 3-PBA as the sole carbon and energy source, but also has a wide range of catabolic abilities. Among the PPs tested, beta-cypermethrin was preferentially transformed by *K. pneumoniae* BPBA052; the respective degradation rates for deltamethrin, permethrin, fenvalerate were 40.57%, 30.86%, and 28.41% within 48 h; and bifenthrin was found to be the most persistent. Zhao et al. (2016) reported that *Bacillus licheniformis* B-1 could degrade approximately 50% β -cypermethrin (100 mg/L) after 72 h, but was unable to degrade 3-PBA; Zhu et al. (2016) reported that *Aspergillus oryzae* M-4 could degrade 80.62% 3-PBA (100 mg/L) within 5 d, but did not report the ability to degrade PPs. In this study, the disappearance rate of 3-PBA and beta-cypermethrin by *K. pneumoniae* BPBA052 was >90% within a much shorter time (2.38 d and 7.27 d, respectively). Compared to *Bacillus licheniformis* B-1 and *Aspergillus oryzae* M-4, *K. pneumoniae* BPBA052 was not only capable of degrading 3-PBA, but could also degrade a series of PPs and benzene compounds. However, it is well known that the microbes in practical application is highly influenced by many factors, including the temperature, substrate concentration, initial pH, nutrients, microorganisms present, and additional carbon sources (Singh et al. 2006; Zhang et al. 2009). Thus, the degradation characteristics of *K. pneumoniae* BPBA052 used against 3-PBA or PPs needs further study.

In our studies of the metabolic pathway of 3-PBA, we did not observe abiotic degradation of 3-PBA in the experimental groups or cell-free controls, and we speculate that the metabolism of 3-PBA was initiated by extremely rapid enzymatic reactions (Chen et al. 2012a). The results of enzyme assay indicated that *K. pneumoniae* BPBA052 can secrete multiple oxidases. Moreover, cloning of *LPH*, *CatA*, and *PRO β* further provided molecular biological evidence that *K. pneumoniae* BPBA052 can degrade 3-PBA metabolites, which is

also consistent with the predictions made by Chen et al. (2012a), Tang et al. (2013), and Zhang et al. (2012). The degradation pathway of 3-PBA indicates that 3-PBA generates phenol (a common toxicant and environmental pollutant) and the metabolite protocatechuate by oxidation and cleavage of the diphenyl ether moiety. It is very important to determine whether harmful metabolites like phenol can be continuously degraded. Our data showed that *K. pneumoniae* BPBA052 can efficiently degrade phenol, catechol, and protocatechuate with high enzyme activity, and successful cloning of the *LPH*, *CatA*, and *PROβ* genes further validated the key enzymatic reactions in the 3-PBA-degradation pathway. Firstly, CatA is a intradiol-cleaving enzyme that catalyzes intradiol-cleavage in which catechol + O₂ become cis,cis-muconate. PRO is a non-heme iron intradiol-cleaving enzyme that can catalyze the oxidative ring cleavage of protocatechuate (i.e., it can break the C₁–C₂ bond and utilize Fe³⁺) to produce carboxy-cis,cis-muconate. Both carboxy-cis, cis-muconate and cis,cis-muconate can further enter Krebs's cycle and finally become converted into H₂O and CO₂. Our findings were used to predict the bio-metabolic mechanism of 3-PBA, and showed that *K. pneumoniae* BPBA052 has a strong 3-PBA metabolic capacity and has the potential to fully mineralize 3-PBA.

5. Conclusions

In this work, we isolated *K. pneumoniae* BPBA052, which has highly efficient 3-PBA- and PPs-degradation abilities. Moreover, the optimum conditions for 3-PBA degradation were identified and a complete mineralization pathway of 3-PBA degradation by oxidation, diaryl ether cleavage, intradiol-cleavage, and entry into Krebs's cycle in the microorganism is proposed. Additionally, this is the first report focused on 3-PBA biodegradation by a bacterial strain from the *Klebsiella* genus. These results suggest that the isolated *K. pneumoniae* BPBA052 has great potential to function as a degrader in eliminating 3-PBA and PPs from the environment.

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Conflicts interest: Jie Tang, Qiong Hu, Bo Liu, Dan Lei, Tingting Chen, Qing Sun, Chaoyi Zeng, and Qing Zhang declare that there are no conflicts of interest.

References

Abraham, J., and Silambarasan, S. 2014. Role of novel fungus *Lasiodiplodia* sp. JAS12 and plant growth promoting bacteria *Klebsiella pneumoniae* JAS8 in mineralization of endosulfan and its metabolites. Ecol. Eng. 70: 235–240. doi: 10.1016/j.ecoleng.2014.05.029

Bradford, M.M. 1976. A rapid and sensitive method for the quantitation of microgram quantities of protein utilizing the principle of protein-dye binding. Anal. Biochem. 72: 248–254.

- doi:10.1016/0003-2697(76)90527-3
- Brenner, D.R., Krieg, N.R., Staley, J.T. 2005. *Bergey's Manual® of Systematic Bacteriology*. Springer.
- doi:10.1007/0-387-29298-5
- Chen, S.H., Hu, M.Y., Liu, J.J., Zhong, G.H., Yang, L., Rizwan-ul-Haq, M., et al. 2011a. Biodegradation of beta-cypermethrin and 3-phenoxybenzoic acid by a novel *Ochrobactrum lupini* DG-S-01. *J. Hazard. Mater.* 187: 433–440. doi: 10.1016/j.jhazmat.2011.01.049
- Chen, S.H., Hu, W., Xiao, Y., Deng, Y.Y., Jia, J.W., Hu, M.Y. 2012a. Degradation of 3-phenoxybenzoic acid by a *Bacillus* sp. *PLoS One* 7: e50456. doi: 10.1371/journal.pone.0050456
- Chen, S.H., Liu, C.L., Peng, C.Y., Liu, H.M., Hu, M.Y., Zhong, G.H. 2012b. Biodegradation of chlorpyrifos and its hydrolysis product 3,5,6-trichloro-2-pyridinol by a new fungal strain *Cladosporium cladosporioides* Hu-01. *PLoS One* 7: e47205. doi: 10.1371/journal.pone.0047205
- Chen, S.H., Yang, L., Hu, M.Y., Liu, J.J. 2011b. Biodegradation of fenvalerate and 3-phenoxybenzoic acid by a novel *Stenotrophomonas* sp. strain ZS-S-01 and its use in bioremediation of contaminated soils. *Appl. Microbiol. Biot.* 90: 755–767. doi: 10.1007/s00253-010-3035-z
- Chen, S.H., Zhang, Y.B., Hu, M.Y., Geng, P., Li, Y.N., An, G.D. 2011c. Bioremediation of β -cypermethrin and 3-phenoxybenzoic acid in soils. *Water Resource and Environmental Protection (ISWREP) International Symposium on Xi'an China*. doi: 10.1109/ISWREP.2011.5893579
- Cycoń, M., and Piotrowska-Seget, Z. 2016. Pyrethroid-degrading microorganisms and their potential for the bioremediation of contaminated soils: a review. *Front. Microbiol.* 7: 1463. doi: 10.3389/fmicb.2016.01463
- Dalal, S., Panigrahi, D.P., Randhawa, G.S., Dubey, R.C. 2012. *CatA* gene in a potential *Corynebacterium* strain is responsible for its efficiency in phenol bioremoval. *Polycycl. Aromat. Comp.* 32: 423–438. doi: 10.1080/10406638.2011.645267

- 386 Halden, R.U., Tepp, S.M., Halden, B.G., Dwyer, D.F. 1999. Degradation of 3-phenoxybenzoic acid in soil by
387 *Pseudomonas pseudoalcaligenes* POB310(pPOB) and two modified *Pseudomonas* strains. Appl. Environ.
388 Microb. 65: 3354–3359. PMCID: PMC91504
- 389 Jin, M.Q., Li, L., Xu, C., Wen, Y.Z., Zhao, M.R. 2010. Estrogenic activities of two synthetic pyrethroids and
390 their metabolites. J. Environ. Sci. 22: 290–296. doi: 10.1016/S1001-0742(09)60107-8
- 391 Kuan, K.B., Othman, R., Abdul, R.K., Shamsuddin, Z.H. 2016. Plant growth-promoting rhizobacteria
392 inoculation to enhance vegetative growth, nitrogen fixation and nitrogen remobilisation of maize under
393 greenhouse conditions. PLoS One 11: e0152478. doi: 10.1371/journal.pone.0152478
- 394 Liu, F.F., Chi, Y.L., Wu, S., Jia, D.Y., Yao, K. 2014. Simultaneous degradation of cypermethrin and its
395 metabolite, 3-phenoxybenzoic acid, by the cooperation of *Bacillus licheniformis* B-1 and *sphingomonas* sp.
396 SC-1. J. Agr. Food Chem. 62: 8256–8262. doi: 10.1021/jf502835n
- 397 Maloney, S.E., Maule, A., Smith, A.R. 1988. Microbial transformation of the pyrethroid insecticides:
398 permethrin, deltamethrin, fastac, fenvalerate, and fluvalinate. Appl. Environ. Microb. 54: 2874–2876.
399 PMCID:PMC204391
- 400 Martínez Vidal, J.L., Plaza-Bolaños, P., Romero-González, R., Garrido Frenich, A. 2009. Determination of
401 pesticide transformation products: a review of extraction and detection methods. J. Chromatogr. A. 1216:
402 6767–6788. doi: 10.1016/j.chroma.2009.08.013
- 403 McCoy, M.R., Yang, Z., Fu, X., Ahn, K.C., Gee, S.J., Bom, D.C., et al. 2012. Monitoring of total type II
404 pyrethroid pesticides in citrus oils and water by converting to a common product 3-phenoxybenzoic acid. J.
405 Agr. Food Chem. 60: 5065–5070. doi: 10.1021/jf2051653
- 406 Milo, R.E., Duffner, F.M., Müller, R. 1999. Catechol 2,3-dioxygenase from the thermophilic, phenol-degrading
407 *Bacillus thermoleovorans* strain A2 has unexpected low thermal stability. Extremophiles 3: 185–190. doi:

- 408 10.1007/s007920050115
- 409 Moreno, M.D.L., Sánchez-Porro, C., Piubeli, F., Frias, L., García, M.T., Mellado, E. 2011. Cloning,
410 characterization and analysis of cat and ben genes from the phenol degrading halophilic bacterium
411 *Halomonas organivorans*. PLoS One 6: e21049. doi:10.1371/journal.pone.0021049
- 412 Ngai, K.L., Neidle, E.L., Ornston, L.N. 1990. Catechol and chlorocatechol 1,2-dioxygenases. Method Enzymol.
413 188: 122–126. doi:10.1016/0076-6879(90)88022-3
- 414 Santos, V.L.D., Monteiro, A.D.S., Braga, D.T., Santoro, M.M. 2009. Phenol degradation by *Aureobasidium*
415 *pullulans* FE13 isolated from industrial effluents. J. Hazard. Mater. 161: 1413–1420. doi:
416 10.1016/j.jhazmat.2008.04.112
- 417 Singh, B.K., Walker, A., Wight, D.J. 2006. Bioremedial potential of fenamiphos and chlorpyrifos degrading
418 isolates: influence of different environmental conditions. Soil Biol. Biochem. 38: 2682–2693. doi:
419 10.1016/j.soilbio.2006.04.019
- 420 Sun, H., Chen, W., Xu, X.L., Ding, Z., Chen, X.D., Wang, X.R. 2014. Pyrethroids and their metabolite,
421 3-phenoxybenzoic acid showed similar (anti)estrogenic activity in human and rat estrogen receptor
422 α -mediated reporter gene assays. Environ. Toxicol. Pharmacol. 37: 371–377. doi:
423 10.1016/j.etap.2013.11.031
- 424 Tallur, P.N., Megadi, V.B., Ninnekar, H.Z, 2008. Biodegradation of cypermethrin by *Micrococcus* sp. strain
425 CPN 1. Biodegradation 19: 77–82. doi:10.1007/s10532-007-9116-8
- 426 Tang, J., Liu, B., Chen, T.T., Yao, K., Zeng, L., Zeng, C.Y., et al. 2018. Screening of a
427 beta-cypermethrin-degrading bacterial strain *Brevibacillus parabrevis* BCP-09 and its biochemical
428 degradation pathway. Biodegradation 1–17. <https://doi.org/10.1007/s10532-018-9850-0>
- 429 Tang, J., Yao, K., Liu, S.L., Jia, D.Y., Chi, Y.L., Zeng, C.Y., et al. 2013. Biodegradation of 3-phenoxybenzoic

- acid by a novel *Sphingomonas* sp. SC-1. Fresenius Environ. Bull. 22: 1564–1572.
- Wang, X., He, B.N., Kong, B.D., Wei, L., Wang, R., Zhou, C.Q., et al. 2017. β -Cypermethrin and its metabolite 3-phenoxybenzoic acid exhibit immunotoxicity in murine macrophages. Acta Biochim Biophys Sin 49: 1083–1091. doi: 10.1093/abbs/gmx111
- Watkins, D.J., Fortenberry, G.Z., Sánchez, B.N., Barr, D.B., Panuwet, P., Schnaas, L., Osorio-Valencia, E., et al. 2016. Urinary 3-phenoxybenzoic acid(3-PBA) levels among pregnant women in Mexico City: distribution and relationships with child neurodevelopment. Environ. Res. 147: 307–313. doi: 10.1016/j.envres.2016.02.025
- Ye, X.Q., Pan, W.Y., Zhao, Y.H., Zhao, S.L., Zhu, Y.M., Liu, W.P., et al. 2016. Association of pyrethroids exposure with onset of puberty in Chinese girls. Environ. Pollut. 227: 606–612. doi: 10.1016/j.envpol.2017.04.035
- Zhang, C., Jia, L., Wang, S.H., Qu, J., Li, K., Xu, L.L., et al. 2010. Biodegradation of beta-cypermethrin by two *Serratia* spp. with different cell surface hydrophobicity. Bioresour Technol. 101: 3423–3429. doi:10.1016/j.biortech.2009.12.083
- Zhang, H., Yang, C., Zhao, Q., Qiao, C.L. 2009. Development of an autofluorescent organophosphates-degrading *Stenotrophomonas* sp. with dehalogenase activity for the biodegradation of hexachlorocyclohexane (HCH). Bioresour Technol. 100: 3199–3204. doi: 10.1016/j.biortech.2009.02.008
- Zhang, J., Lang, Z.F., Zheng, J.W., Hang, B.J., Duan, X.Q., He, J., et al. 2012. *Sphingobium jiangsuense* sp. nov., a 3-phenoxybenzoic acid-degrading bacterium isolated from a wastewater treatment system. Int. J. Syst. Evol. Micr. 62: 800–805. doi: 10.1099/ijs.0.029827-0
- Zhao, J.Y., Chi, Y.L., Xu, Y.C., Jia, D.Y., Yao, k. 2016. Co-metabolic degradation of β -cypermethrin and 3-phenoxybenzoic acid by co-culture of *Bacillus licheniformis* B-1 and *Aspergillus oryzae* M-4. PLoS One

- 452 11: e0166796. doi:10.1371/journal.pone.0.66796
- 453 Zhu, Y.T., Li, J.L., Yao, K., Zhao, N., Zhou, K., Hu, X.J., et al. 2016. Degradation of 3-phenoxybenzoic acid by
- 454 a filamentous fungus *Aspergillus oryzae* M-4 strain with self-protection transformation. Appl. Microbiol.
- 455 Biot. 100: 9773–9786. doi: 10.1007/s00253-016-7847-3

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Legends to Figures

Fig. 1. (a) Scanning electron microscopic morphology. (b) Phylogenetic analysis of *K. pneumoniae* BPBA052, based on the 16S rRNA sequence. The numbers shown in parentheses are GenBank sequence accession numbers. The numbers at the nodes indicate bootstrap values from the neighbor-joining analysis of 1,000 resampled data sets. The bar represents sequence divergence.

Fig. 2. Growth and 3-PBA degradation curve of *K. pneumoniae* BPBA052.

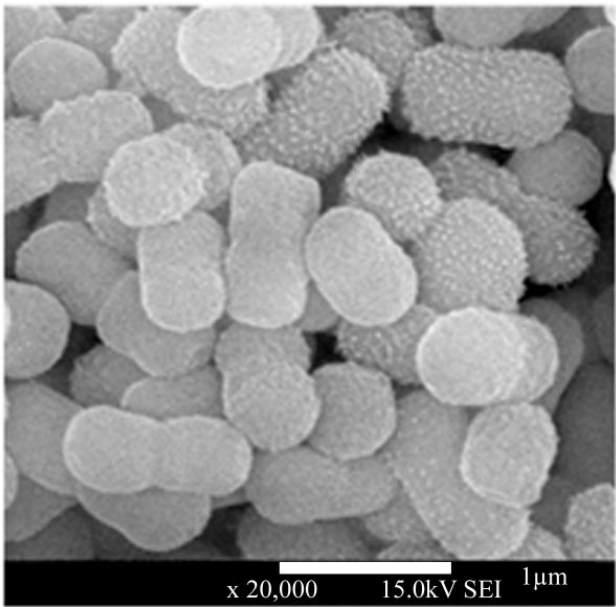
Fig. 3. Degradation curves for 3-PBA under different culture conditions by *K. pneumoniae* BPBA052. Variations were made in the (A) culture temperature, (B) 3-PBA concentration, and (C) initial pH.

Fig. 4. Response surface plot showing the effects of temperature and initial pH on 3-PBA degradation by *K. pneumoniae* BPBA052 with a 3-PBA concentration of 150 mg/L.

Fig. 5. (a) Degradation of different metabolites during 3-PBA degradation by *K. pneumoniae* BPBA052. (b) Biodegradation metabolic pathway of 3-PBA by *K. pneumoniae* BPBA052.

Legends to Figures

a



b

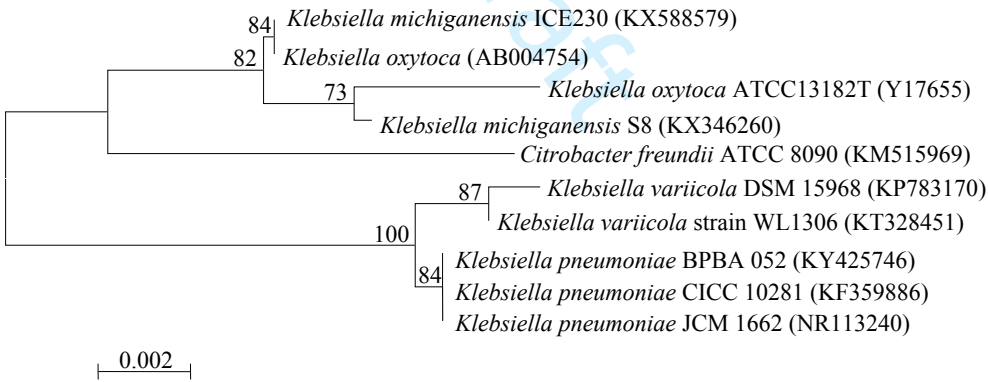


Fig. 1

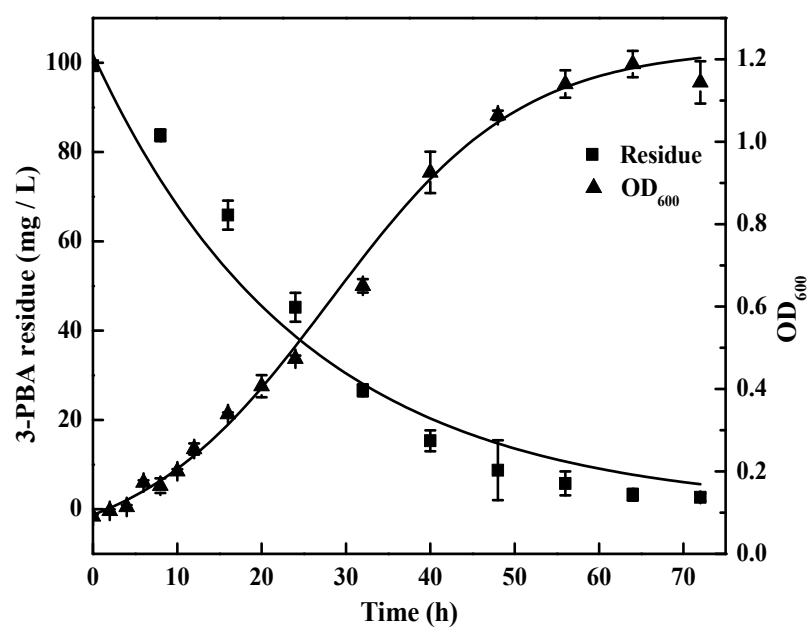
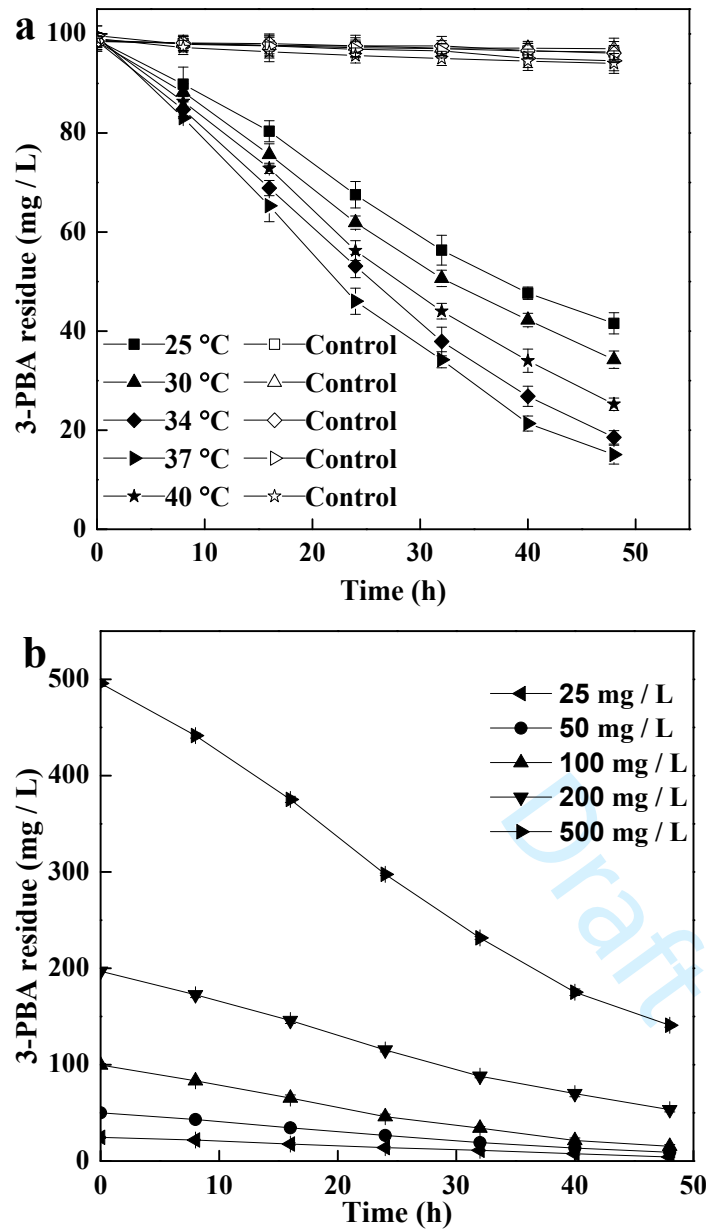


Fig. 2

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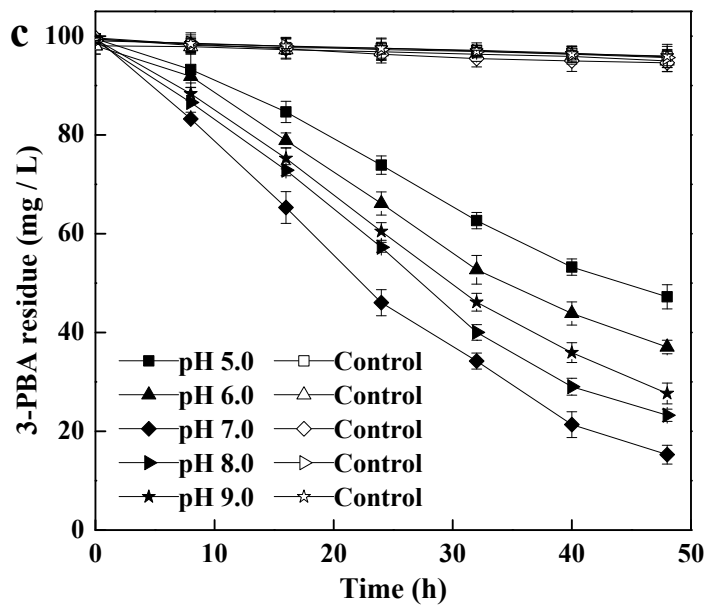
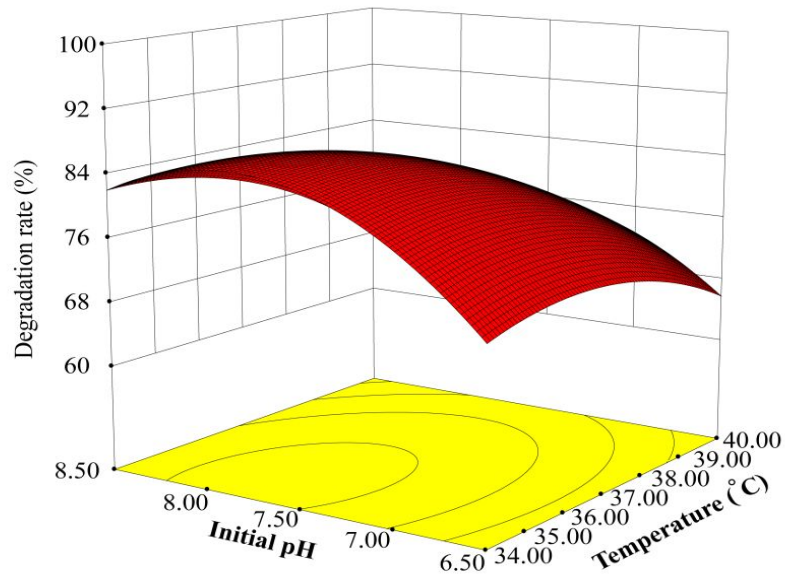


Fig. 3

**Fig. 4**

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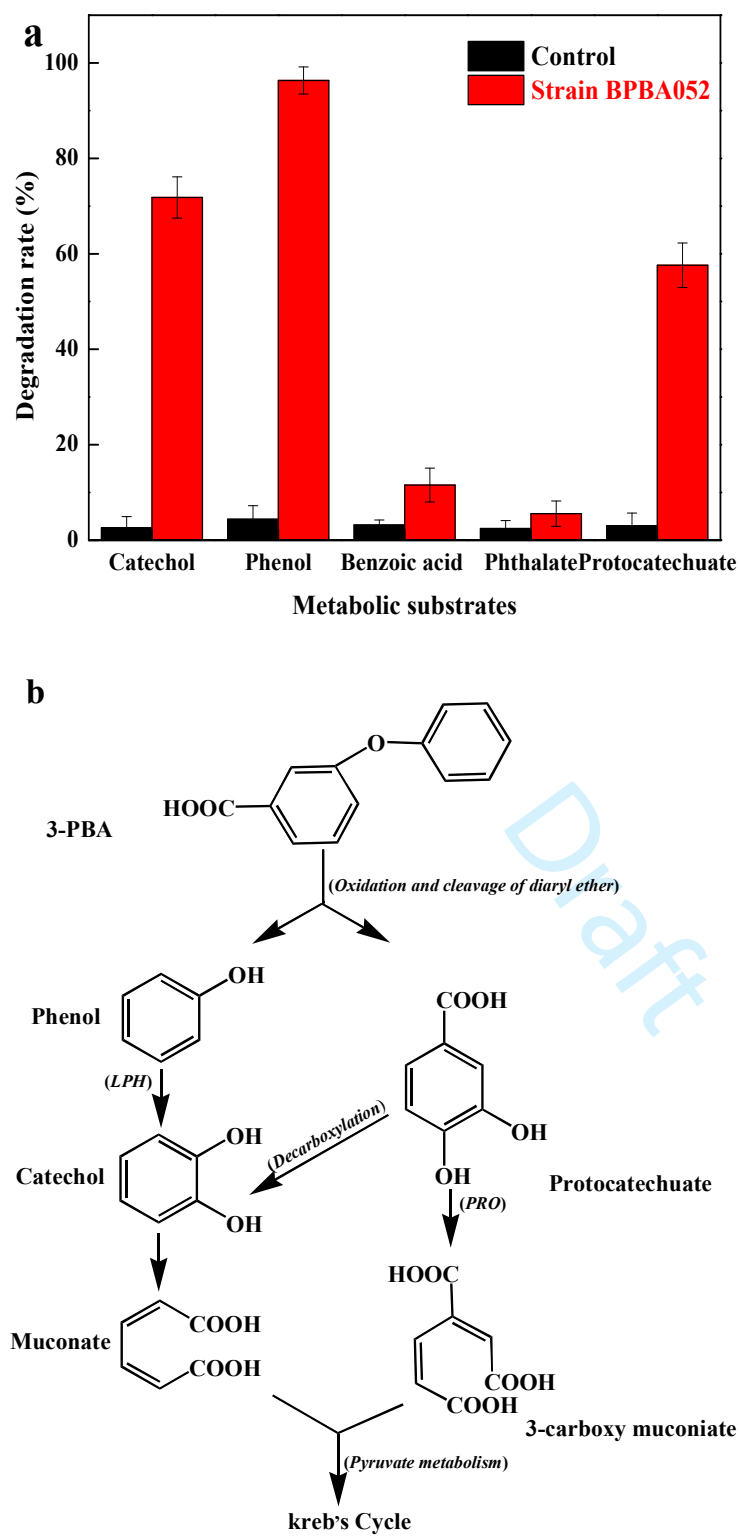


Fig. 5

Table 1. Box–Behnken experimental design matrix with 3 dependent variables for 3-PBA degradation

Run	X_1 (°C)	X_2 (mg/L)	X_3	Degradation rate (%)
1	0	0	0	82.25
2	0	−1	−1	70.44
3	0	0	0	85.21
4	−1	0	1	81.64
5	0	1	−1	73.89
6	1	1	0	76.56
7	−1	0	−1	70.15
8	0	−1	1	78.09
9	0	0	0	84.96
10	0	0	0	85.38
11	−1	1	0	87.11
12	−1	−1	0	84.25
13	1	0	−1	66.26
14	1	−1	0	74.55
15	0	0	0	85.88
16	1	0	1	68.56
17	0	1	1	80.56

X_1 refers to temperature: −1 (34°C), 0 (37°C), +1 (40°C); X_2 refers to the 3-PBA concentration: −1 (50 mg/L), 0 (100 mg/L), +1 (150 mg/L); and X_3 refers to the initial pH: −1 (6.5), 0 (7.5), +1 (8.5). The data are presented as the mean of three replicates with the standard deviation, which was within 5% of the mean.

Table 2. ANOVA for the response surface quadratic model for 3-PBA degradation

Source	Sum of squares	Degrees of freedom	Mean square	F value	p-value
Model	738.22	9	82.02	59.22	< 0.0001
X_1	173.17	1	173.17	125.01	< 0.0001
X_2	14.55	1	14.55	10.51	0.0142
X_3	98.77	1	98.77	71.31	< 0.0001
X_1X_2	0.18	1	0.18	0.13	0.7287
X_1X_3	21.11	1	21.11	15.24	0.0059
X_2X_3	0.24	1	0.24	0.17	0.6896
X_1^2	70.97	1	70.97	51.23	0.0002
X_2^2	7.12E-04	1	7.12E-04	5.14E-04	0.9825
X_3^2	339.39	1	339.39	245.01	< 0.0001
Residual	9.70	7	1.39		
Lack of fit	1.52	3	0.51	0.25	0.8597
Pure error	8.18	4	2.04		
Total	747.92	16			

$R^2 = 0.9870$; adjusted $R^2 = 0.9704$; coefficient of variation (CV) = 1.5%.

Table 3. Kinetic parameters for the degradation of different substrates by *K. pneumoniae* BPBA052

Substrate	Equation	k (h ⁻¹)	$t_{1/2}$ (h)	R ²
Beta-cypermethrin	$C_t = 100.6022 \times e^{-0.0132t}$	$k = 0.0132$	$t_{1/2} = 52.51$	0.9965
Permethrin	$C_t = 101.2910 \times e^{-0.0083t}$	$k = 0.0083$	$t_{1/2} = 83.51$	0.9792
Bifenthrin	$C_t = 100.7825 \times e^{-0.0067t}$	$k = 0.0067$	$t_{1/2} = 103.45$	0.9928
Deltamethrin	$C_t = 100.6814 \times e^{-0.0111t}$	$k = 0.0111$	$t_{1/2} = 62.45$	0.9965
Fenvalerate	$C_t = 99.5318 \times e^{-0.0091t}$	$k = 0.0091$	$t_{1/2} = 76.17$	0.9922

Notes: C_t is the residual concentration of PPs at time t (mg/L), t and k are respectively the degradation period (h) and the degradation rate constant (h⁻¹), $t_{1/2}$ is the half-life of PPs, and R² is the correlation coefficient.

Table 1. Degenerate primers used in this study

Primer name	Sequence (5'→3')
<i>PROβ</i> -f	ATGAAYRAHAAATGGTCMCCRC
<i>PROβ</i> -r	TCATYRRRMGGCCCCCTG
<i>PROα</i> -f	ATGAAAGAYYAYYYASCYGAAACC
<i>PROα</i> -r	TYARCGATCGAAGAASACGGT
<i>CatA</i> -f	ATGAACCGTCAACAAATTGAT
<i>CatA</i> -r	TTAAGCRCTAGCGCGACGT
<i>CatB</i> -f	ATGAGTAATGYWYYCGKACAACAR
<i>CatB</i> -r	TTACTCTTTGGCGCRYASR
<i>LPH</i> -f	AGGCATCAAGATCACCGACTG
<i>LPH</i> -r	CGCCAGAACCATTTATCGATC

Notes: R = A/G, Y = C/T, M = A/C, K = G/T, S = C/G, W = A/T